

Chapter 4: Sturm–Liouville Theory

Mathematical Physics

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Outline

- 1 Review of linear algebra
- 2 Sturm–Liouville operators
- 3 Green's function on an interval

Where we are going

This chapter builds the operator picture behind the second-order equations of later chapters, using only basic calculus and inner products of functions.

- Functions act as vectors in an inner-product space; Fourier coefficients are extracted by orthogonality.
- A Sturm–Liouville eigenvalue problem is written, with this chapter's sign convention, as

$$\frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + q(x)u(x) = -\lambda w(x)u(x),$$

where λ is the eigenvalue and typically $p(x) > 0$ and $w(x) > 0$ in the interior of the interval.

- Integration by parts identifies the *formal adjoint*. Boundary conditions that kill the boundary form give the symmetry used below.
- Green's identity then yields real eigenvalues and weighted orthogonality of eigenfunctions.
- Legendre and Bessel (drum) modes illustrate the theory. A real second-order equation can be put into divergence form by an integrating factor on an interval where its leading coefficient is nonzero.
- A point source then defines the Green function.

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Vector spaces and bases

A *vector space* is a collection of objects that is closed under addition and scalar multiplication. Thus, if u and v belong to the space and a, b are scalars, then

$$au + bv$$

also belongs to the space. The vectors may be functions rather than geometric arrows.

Examples.

- The polynomials of degree at most two, $a_0 + a_1x + a_2x^2$, form a three-dimensional vector space.
- The smooth functions on $[0, 1]$ satisfying $u(0) = u(1) = 0$ form a vector space, because

$$(au + bv)(0) = au(0) + bv(0) = 0, \quad (au + bv)(1) = au(1) + bv(1) = 0.$$

A *basis* is a set of vectors from which every vector in the space has a unique linear combination. Existence is called *spanning*; uniqueness follows from *linear independence* (only zero coefficients give the zero vector). For the quadratic polynomials, $\{1, x, x^2\}$ is a basis, and

$$1 + 2x + 3x^2 = 1 \cdot 1 + 2 \cdot x + 3 \cdot x^2. \quad (4.1.1)$$

The number of elements in a finite basis is the *dimension*. To discuss infinite bases, we first need a notion of distance and convergence.

Inner products, norms, and coefficients

Step 1: choose the inner product. We use the convention that the inner product is *linear in its first slot*. An inner product $(\cdot, \cdot)$ satisfies

- ❶ $(au + bv, w) = a(u, w) + b(v, w),$
- ❷ $(u, v) = \overline{(v, u)},$
- ❸ $(u, u) \geq 0,$ with equality only for the zero vector.

In particular, for a scalar $c,$

$$(cu, v) = c(u, v). \quad (4.1.2)$$

Conjugate symmetry forces *conjugate-linearity in the second slot*:

$$(u, cv) = \overline{c}(u, v). \quad (4.1.3)$$

For example, with $u = (1, 0)$ and $(u, v) = u_1 \overline{v_1} + u_2 \overline{v_2},$

$$(iu, u) = i, \quad (u, iu) = -i.$$

The conjugate is essential: $(iu, iu) = 1,$ not $-1.$

The standard L^2 inner product on an interval $[a, b]$ is

$$(u, v) = \int_a^b u(x) \overline{v(x)} dx. \quad (4.1.4)$$

Inner products, norms, and coefficients (cont.)

If w is a real weight with $w(x) > 0$ on the interior (it may vanish at isolated endpoints), the weighted inner product is

$$(u, v)_w = \int_a^b u(x) \overline{v(x)} w(x) dx. \quad (4.1.5)$$

For example, the Bessel radial weight on $[0, 1]$ is $w(s) = s$. Then

$$(u, u)_w = \int_a^b |u(x)|^2 w(x) dx$$

is positive for every nonzero continuous u of interest in this course.

The norm induced by an inner product is

$$\|u\| = \sqrt{(u, u)}, \quad \|u\|_w = \sqrt{(u, u)_w}. \quad (4.1.6)$$

A nonzero function is normalized by replacing it with $u/\|u\|$.

Two vectors are *orthogonal* when $(u, v) = 0$. An orthonormal set $\{u_i\}$ satisfies

$$(u_i, u_j) = \delta_{ij}, \quad (4.1.7)$$

where $\delta_{ij} = 1$ for $i = j$ and 0 otherwise.

Inner products, norms, and coefficients (cont.)

Step 2: extract coefficients by orthogonality. Suppose first that

$$f = \sum_{i=1}^N a_i u_i$$

in an orthonormal basis. Taking the inner product with u_j gives

$$\begin{aligned}(f, u_j) &= \left(\sum_{i=1}^N a_i u_i, u_j \right) \\ &= \sum_{i=1}^N a_i (u_i, u_j) \\ &= \sum_{i=1}^N a_i \delta_{ij} \\ &= a_j.\end{aligned}\tag{4.1.8}$$

(The same formula holds for an infinite expansion once one knows the series converges in the L^2 norm, so the inner product may pass under the limit.)

Inner products, norms, and coefficients (cont.)

If the basis is orthogonal but not normalized, then

$$(f, u_j) = \sum_i a_i (u_i, u_j) = a_j (u_j, u_j) = a_j \|u_j\|^2,$$

so

$$a_j = \frac{(f, u_j)}{\|u_j\|^2}. \quad (4.1.9)$$

Infinite-dimensional bases and L^2

For an infinite series of functions, we must say how its partial sums approach the function. In $L^2(0, 1)$ the distance is measured by the square integral:

$$\|f\|_2 = \left(\int_0^1 |f(x)|^2 dx \right)^{1/2} < \infty.$$

Functions differing only at finitely many points, or on a set that does not affect integrals, represent the same L^2 vector. No measure theory is needed here.

Use the sine modes

$$u_n(x) = \sin(n\pi x), \quad n = 1, 2, \dots \quad (4.1.10)$$

The next calculation shows that they are orthogonal, with $\|u_n\|^2 = 1/2$. Orthogonality alone is not enough: removing u_1 leaves an orthogonal set that cannot reproduce u_1 . The additional property is called *completeness*.

Infinite-dimensional bases and L^2 (cont.)

Completeness of the sine system (quoted). Every $f \in L^2(0, 1)$ has the expansion

$$f = \sum_{n=1}^{\infty} a_n u_n, \quad a_n = \frac{(f, u_n)}{\|u_n\|^2}, \quad (4.1.11)$$

with mean-square convergence:

$$\left\| f - \sum_{n=1}^N a_n u_n \right\|_2 \longrightarrow 0.$$

This is the infinite-series version of the projection formula (4.1.9). Its proof of completeness belongs to Fourier analysis; we will compute the norm and use the coefficients.

Mean-square convergence concerns the integrated squared error, not the value at every point. Every sine sum is zero at 0 and 1, even when f is not.

Example: orthogonality of the sine modes

Let $u_m(x) = \sin(m\pi x)$ and $u_n(x) = \sin(n\pi x)$, with m, n positive integers. Use the product-to-sum identity

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]. \quad (4.1.12)$$

For $m \neq n$, integration gives

$$\begin{aligned} \int_0^1 u_m u_n dx &= \left[\frac{\sin((m-n)\pi x)}{2(m-n)\pi} - \frac{\sin((m+n)\pi x)}{2(m+n)\pi} \right]_0^1 \\ &= 0, \end{aligned} \quad (4.1.13)$$

because sine is zero at every integer multiple of π .

For $m = n$, use $\sin^2(m\pi x) = [1 - \cos(2m\pi x)]/2$ instead:

$$\int_0^1 u_m^2 dx = \left[\frac{x}{2} - \frac{\sin(2m\pi x)}{4m\pi} \right]_0^1 = \frac{1}{2}. \quad (4.1.14)$$

Together,

$$(u_m, u_n) = \frac{1}{2} \delta_{mn}. \quad (4.1.15)$$

Thus multiplying each u_n by $\sqrt{2}$ would give a norm-one basis.

Example: Fourier sine-series coefficients

For $f \in L^2(0, 1)$, write

$$f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \quad (4.1.16)$$

in the mean-square sense just defined. The projection formula (4.1.9) and the norm $\|u_k\|^2 = 1/2$ give

$$\begin{aligned} a_k &= \frac{(f, u_k)}{\|u_k\|^2} = \frac{\int_0^1 f(x) \sin(k\pi x) dx}{1/2} \\ &= 2 \int_0^1 f(x) \sin(k\pi x) dx. \end{aligned} \quad (4.1.17)$$

The factor 2 comes from the norm, not from an additional convention. The same projection rule will give coefficients for Bessel and polynomial expansions; only the interval, weight, and norm will change.

Linear operators and formal adjoints

A linear operator acts on functions. It satisfies $L(au + bv) = aLu + bLv$. Differentiation, multiplication by a fixed function, and $Au(x) = \int_{x_0}^x u(t) dt$ are examples. The allowed functions must also satisfy any specified boundary conditions.

Integration by parts moves a derivative. With $D = d/dx$,

$$\begin{aligned}(u, Dv) &= \int_0^1 u \overline{v'} dx \\ &= [u\overline{v}]_0^1 - \int_0^1 u' \overline{v} dx = [u\overline{v}]_0^1 + (-Du, v).\end{aligned}\tag{4.1.18}$$

The expression left behind is the *formal adjoint*: $D^\dagger = -D$. Even if the boundary term vanishes, a minus sign remains. Thus D is not formally self-adjoint.

Linear operators and formal adjoints (cont.)

Two integrations restore the sign. For $L = D^2$,

$$\begin{aligned}(u, Lv) &= \int_0^1 u \overline{v''} dx \\ &= [u \overline{v'}]_0^1 - \int_0^1 u' \overline{v'} dx\end{aligned}\tag{4.1.19}$$

$$= [u \overline{v'} - u' \overline{v}]_0^1 + \int_0^1 u'' \overline{v} dx.\tag{4.1.20}$$

Hence $L^\dagger = L$. If $u(0) = u(1) = v(0) = v(1) = 0$, both boundary contributions vanish:

$$(u, Lv) = (Lu, v).\tag{4.1.21}$$

This symmetry is the property that makes eigenfunctions with different eigenvalues orthogonal.

Both the expression and the boundary conditions matter. On a regular interval, homogeneous Dirichlet (zero value), Neumann (zero derivative), or real Robin (zero linear combination) conditions make this expression a self-adjoint problem. Singular endpoints require a separate limiting boundary check.

Eigenfunctions and the orthogonality theorem

An eigenpair (u, λ) of an operator L satisfies

$$Lu = \lambda u, \quad u \neq 0. \quad (4.1.22)$$

A generalized eigenvalue problem introduces a prescribed real weight $w(x)$:

$$Lu = \lambda wu. \quad (4.1.23)$$

In the Sturm–Liouville convention used later, the right-hand side is $-\lambda wu$.

Examples.

- On a domain containing $e^{\lambda x}$ without incompatible boundary conditions,

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}.$$

- For real $m \neq 0$, the functions $\cos(mx)$ and $\sin(mx)$ are eigenfunctions of the second-derivative differential expression with eigenvalue $-m^2$.

Eigenfunctions and the orthogonality theorem (cont.)

Theorem

Every eigenvalue of a self-adjoint operator is real. Eigenfunctions belonging to distinct eigenvalues are orthogonal.

Proof. Let $Lu_i = \lambda_i u_i$ and $Lu_j = \lambda_j u_j$. Self-adjointness and conjugate-linearity in the second slot give

$$\begin{aligned}\lambda_i(u_i, u_j) &= (Lu_i, u_j) \\ &= (u_i, Lu_j) \\ &= (u_i, \lambda_j u_j) = \overline{\lambda_j}(u_i, u_j).\end{aligned}$$

Therefore

$$(\lambda_i - \overline{\lambda_j})(u_i, u_j) = 0.$$

Taking $j = i$ and using $(u_i, u_i) > 0$ shows that $\lambda_i = \overline{\lambda_i}$, so every eigenvalue is real. If $\lambda_i \neq \lambda_j$, both eigenvalues are now known to be real, and the same identity forces $(u_i, u_j) = 0$. The Sturm–Liouville proof below repeats this calculation with the weight and boundary term visible.

Distinct eigenvalues are required for orthogonality. For a repeated eigenvalue, orthogonal linear combinations can be chosen within its solution space.

Eigenfunctions and the orthogonality theorem (cont.)

Application: the fixed string. Let $Lu = u''$ act on twice continuously differentiable functions on $[0, 1]$ that satisfy the Dirichlet conditions $u(0) = u(1) = 0$. First, every eigenvalue is negative. If $Lu = \mu u$ with $u \neq 0$, then

$$\begin{aligned}\mu \|u\|^2 &= (Lu, u) = \int_0^1 u'' \bar{u} \, dx \\ &= [u' \bar{u}]_0^1 - \int_0^1 |u'|^2 \, dx \\ &= - \int_0^1 |u'|^2 \, dx < 0.\end{aligned}$$

(The endpoint term vanishes because $u(0) = u(1) = 0$. The strict inequality holds because $u' = 0$ would make u constant, and the two Dirichlet conditions would then force $u = 0$.) Hence write $\mu = -k^2$ with $k > 0$. The eigenvalue equation becomes

$$u'' + k^2 u = 0, \quad u(0) = u(1) = 0.$$

Its general solution is

$$u(x) = A \cos(kx) + B \sin(kx).$$

Eigenfunctions and the orthogonality theorem (cont.)

The first boundary condition gives

$$0 = u(0) = A \cos 0 + B \sin 0 = A,$$

so $A = 0$. The second gives

$$0 = u(1) = B \sin k.$$

A nonzero eigenfunction requires $B \neq 0$, hence

$$\sin k = 0 \implies k = n\pi, \quad n = 1, 2, 3, \dots$$

Thus

$$u_n(x) = \sin(n\pi x), \quad Lu_n = -(n\pi)^2 u_n. \quad (4.1.24)$$

Because $(u, Lv) = (Lu, v)$ holds under these boundary conditions, the theorem gives $(u_m, u_n) = 0$ for $m \neq n$, in agreement with the direct integral calculation in (4.1.15).

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Formal adjoint of a second-order expression

For real, sufficiently differentiable coefficients, write

$$\ell[u] = p_0 u'' + p_1 u' + p_2 u. \quad (4.2.1)$$

To obtain the formal adjoint, integrate the u'' term twice and the u' term once. With the inner product (4.1.4),

$$\begin{aligned} (\ell[u], v) &= \int_a^b (p_0 u'' + p_1 u' + p_2 u) \bar{v} \, dx \\ &= [p_0 u' \bar{v} - u(p_0 \bar{v})' + p_1 u \bar{v}]_a^b \\ &\quad + \int_a^b u [(p_0 \bar{v})'' - (p_1 \bar{v})' + p_2 \bar{v}] \, dx. \end{aligned} \quad (4.2.2)$$

Because the coefficients are real, the last integrand is $\overline{u [(p_0 v)'' - (p_1 v)' + p_2 v]}$. Therefore

$$\ell^\dagger[v] = (p_0 v)'' - (p_1 v)' + p_2 v. \quad (4.2.3)$$

Expanding the derivatives makes the coefficient comparison explicit:

$$\ell^\dagger[v] = p_0 v'' + (2p_0' - p_1) v' + (p_0'' - p_1' + p_2) v.$$

Formal adjoint of a second-order expression (cont.)

Equality with $\ell[v] = p_0 v'' + p_1 v' + p_2 v$ first requires $2p'_0 - p_1 = p_1$, or $p_1 = p'_0$. Differentiating this relation also gives $p''_0 - p'_1 = 0$, so the coefficient of v agrees as well. Conversely, this single relation makes every coefficient agree. Hence

$$\ell^\dagger = \ell \iff p_1 = p'_0. \quad (4.2.4)$$

The practical test is the product rule:

$$p_0 u'' + p'_0 u' = (p_0 u')'.$$

Under the same condition, the endpoint term in (4.2.2) reduces to

$$\left[p_0 (u' \bar{v} - u \bar{v}') \right]_a^b.$$

Thus the differential expressions agree formally, but the boundary term must still vanish on the allowed functions. We now derive that boundary identity in a form suited to the eigenvalue problem.

The Sturm–Liouville expression and Green's identity

Step 1: put the expression in divergence form. When $p_1 = p_0'$, the first two terms in (4.2.1) combine by the product rule:

$$\begin{aligned}(p_0 u')' &= p_0' u' + p_0 u'' \\ &= p_1 u' + p_0 u''.\end{aligned}$$

Writing $p = p_0$ and $q = p_2$ gives the Sturm–Liouville differential expression

$$\ell[u] = (pu')' + qu. \quad (4.2.5)$$

Step 2: derive Green's identity. Because p and q are real and complex conjugation commutes with differentiation, $\overline{\ell[v]} = \overline{(pv')' + qv} = (p\overline{v'})' + q\overline{v}$. Integrating by parts once in each of the following expressions,

$$\begin{aligned}(u, \ell[v]) &= \int_a^b u [(p\overline{v'})' + q\overline{v}] dx \\ &= [up\overline{v'}]_a^b - \int_a^b pu' \overline{v'} dx + \int_a^b qu\overline{v} dx,\end{aligned} \quad (4.2.6)$$

The Sturm–Liouville expression and Green's identity (cont.)

and

$$\begin{aligned}(\ell[u], v) &= \int_a^b [(pu')' + qu] \bar{v} \, dx \\ &= [pu'\bar{v}]_a^b - \int_a^b pu'\bar{v}' \, dx + \int_a^b qu\bar{v} \, dx.\end{aligned}\tag{4.2.7}$$

Subtracting cancels both integrals and yields Green's identity, also called Lagrange's identity:

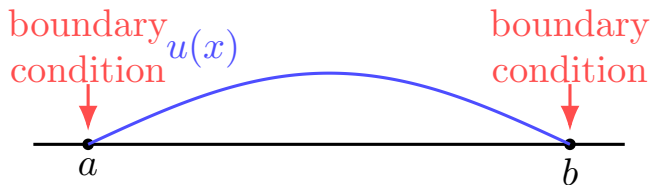
$$(u, \ell[v]) - (\ell[u], v) = [p(u\bar{v}' - u'\bar{v})]_a^b.\tag{4.2.8}$$

The endpoint expression

$$\mathcal{B}[u, v](x) = p(x) (u(x)\overline{v'(x)} - u'(x)\overline{v(x)})\tag{4.2.9}$$

is the Lagrange boundary form.

The Sturm–Liouville expression and Green's identity (cont.)



$$p(u\overline{v}' - u'\overline{v})\Big|_a^b$$

self-adjoint boundary conditions make
the endpoint contribution zero

The Sturm–Liouville expression and Green's identity (cont.)

Step 3: choose boundary conditions that kill the boundary form. For a regular finite interval, standard self-adjoint boundary conditions include the following.

Separated real Robin conditions. At each endpoint impose

$$\alpha_a u(a) + \beta_a (pu')(a) = 0, \quad \alpha_b u(b) + \beta_b (pu')(b) = 0, \quad (4.2.10)$$

where the coefficients are real and $(\alpha, \beta) \neq (0, 0)$ at each endpoint. The same conditions are imposed on v .

To verify cancellation at one endpoint, suppose first that $\beta \neq 0$. Then

$$pu' = -\frac{\alpha}{\beta} u, \quad p\overline{v'} = -\frac{\alpha}{\beta} \overline{v},$$

so

$$\begin{aligned} p(u\overline{v'} - u'\overline{v}) &= u(p\overline{v'}) - (pu')\overline{v} \\ &= u\left(-\frac{\alpha}{\beta}\overline{v}\right) - \left(-\frac{\alpha}{\beta}u\right)\overline{v} \\ &= 0. \end{aligned}$$

The Sturm–Liouville expression and Green’s identity (cont.)

If $\beta = 0$, then $u = v = 0$ at that endpoint (Dirichlet). If $\alpha = 0$, then $pu' = pv' = 0$ there (Neumann or natural boundary condition). Thus (4.2.10) contains Dirichlet and Neumann conditions as special cases.

Periodic conditions. The coupled conditions

$$u(a) = u(b), \quad (pu')(a) = (pu')(b) \quad (4.2.11)$$

and the same conditions for v make $\mathcal{B}[u, v](b) = \mathcal{B}[u, v](a)$, so the difference in (4.2.8) is zero.

Singular endpoints. If p vanishes at an endpoint or the interval is infinite, the bare statement “ $p = 0$ ” is not enough by itself. One checks the limiting boundary form and keeps only those modes for which

$$\lim_{x \rightarrow a^+} \mathcal{B}[u, v](x) = 0, \quad \lim_{x \rightarrow b^-} \mathcal{B}[u, v](x) = 0. \quad (4.2.12)$$

For Legendre polynomials, $p = 1 - x^2$ tends to zero while the polynomials and their derivatives stay finite. For Hermite polynomials, $p = e^{-x^2}$ kills every polynomial Wronskian as $x \rightarrow \pm\infty$: the modes are real, so $u\overline{v}' - u'\overline{v} = uv' - u'v = W[u, v]$, and $\mathcal{B} = e^{-x^2} W[u, v] \rightarrow 0$ because the Gaussian decays faster than any polynomial grows.

The Sturm–Liouville expression and Green's identity (cont.)

When these boundary conditions or verified singular limits make the boundary form vanish, $(u, \ell[v]) = (\ell[u], v)$. The next argument also requires finite weighted squared norms; it establishes reality and orthogonality, not a general self-adjointness or completeness theorem.

The Sturm–Liouville eigenvalue problem

Step 1: write the two eigenvalue equations. With the sign convention of this chapter, the Sturm–Liouville eigenvalue equation is

$$(pu')' + qu = -\lambda wu, \quad (4.2.13)$$

Here p is continuously differentiable and q, w are continuous in the interior; all three are real, with $p > 0$ and $w > 0$ there. We use nonzero eigenfunctions with finite weighted squared norm, $0 < \int_a^b w|u|^2 dx < \infty$. At a singular endpoint, take limits from inside the interval. The chosen boundary conditions must make the limiting Lagrange boundary form vanish.

Let u_i and u_j satisfy

$$(pu'_i)' + qu_i = -\lambda_i wu_i, \quad (4.2.14)$$

$$(pu'_j)' + qu_j = -\lambda_j wu_j. \quad (4.2.15)$$

Conjugating (4.2.15), using reality of p, q, w , gives

$$(p\overline{u'_j})' + q\overline{u_j} = -\overline{\lambda_j} w\overline{u_j}. \quad (4.2.16)$$

The Sturm–Liouville eigenvalue problem (cont.)

Multiply (4.2.14) by $\overline{u_j}$ and (4.2.16) by u_i :

$$\begin{aligned}\overline{u_j}(pu_i')' + qu_i\overline{u_j} &= -\lambda_i wu_i\overline{u_j}, \\ u_i(p\overline{u_j}')' + qu_i\overline{u_j} &= -\overline{\lambda_j} wu_i\overline{u_j}.\end{aligned}$$

Subtracting the second equation from the first cancels the q terms:

$$\overline{u_j}(pu_i')' - u_i(p\overline{u_j}')' = -(\lambda_i - \overline{\lambda_j})wu_i\overline{u_j}. \quad (4.2.17)$$

Step 2: turn the difference into a boundary term. The left-hand side is an exact derivative. Indeed,

$$\begin{aligned}\frac{d}{dx} [p(\overline{u_j}u_i' - u_i\overline{u_j}')] &= \overline{u_j}(pu_i')' + p\overline{u_j}'u_i' - u_i(p\overline{u_j}')' - pu_i'\overline{u_j}' \\ &= \overline{u_j}(pu_i')' - u_i(p\overline{u_j}')'.\end{aligned}$$

Thus (4.2.17) is

$$\frac{d}{dx} [p(\overline{u_j}u_i' - u_i\overline{u_j}')] = -(\lambda_i - \overline{\lambda_j})wu_i\overline{u_j}. \quad (4.2.18)$$

The Sturm–Liouville eigenvalue problem (cont.)

Integrating from a to b gives

$$\left[p(\overline{u_j} u_i' - u_i \overline{u_j}') \right]_a^b = -(\lambda_i - \overline{\lambda_j}) \int_a^b w u_i \overline{u_j} dx. \quad (4.2.19)$$

The integrand in the bracket is $p(\overline{u_j} u_i' - u_i \overline{u_j}') = -\mathcal{B}[u_i, u_j](x)$, the negative of the Lagrange boundary form (4.2.9). Thus the separated Robin conditions (4.2.10), periodic conditions (4.2.11), or verified singular limits (4.2.12) make the left-hand side zero.

Step 3: conclude reality and weighted orthogonality. Set $j = i$. Since $u_i \neq 0$ and $w > 0$ on the interior,

$$0 < \int_a^b w |u_i|^2 dx < \infty.$$

Equation (4.2.19) therefore gives

$$\lambda_i - \overline{\lambda_i} = 0,$$

The Sturm–Liouville eigenvalue problem (cont.)

so every eigenvalue is real. For two distinct eigenvalues, reality gives $\overline{\lambda_j} = \lambda_j$; substituting this into (4.2.19) and using that the boundary conditions make the left-hand bracket vanish leaves

$$0 = -(\lambda_i - \lambda_j) \int_a^b w u_i \overline{u_j} dx,$$
$$\lambda_i \neq \lambda_j \implies \int_a^b w u_i \overline{u_j} dx = 0.$$

Hence

$$(u_i, u_j)_w = 0 \quad (\lambda_i \neq \lambda_j). \quad (4.2.20)$$

Equivalently, $A = -w^{-1}\ell$ satisfies $(Au, v)_w = (u, Av)_w$ under these boundary conditions.

This is the weighted symmetry used in the calculation.

Example: Legendre equation

For a nonnegative integer n , the Legendre equation is

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0, \quad -1 < x < 1. \quad (4.2.21)$$

By the product rule,

$$\begin{aligned} \frac{d}{dx} [(1 - x^2)y'] &= (1 - x^2)y'' + (1 - x^2)'y' \\ &= (1 - x^2)y'' - 2xy'. \end{aligned}$$

Therefore (4.2.21) is equivalent to

$$\frac{d}{dx} [(1 - x^2)y'] = -n(n + 1)y. \quad (4.2.22)$$

The Sturm–Liouville data are

$$p(x) = 1 - x^2, \quad q(x) = 0, \quad w(x) = 1, \quad \lambda_n = n(n + 1).$$

The bounded solution is the Legendre polynomial P_n , normalized by $P_n(1) = 1$. The first examples are

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1).$$

Example: Legendre equation (cont.)

They can be checked directly in the equation. Chapter 6 constructs the whole family by series and Rodrigues' formula. Here we use their polynomial nature to examine the endpoint term: both P_n and P'_n stay finite at $x = \pm 1$.

For two degrees $n \neq k$, the eigenvalues $n(n+1)$ and $k(k+1)$ differ. The endpoint term is

$$(1-x^2)(P_n P'_k - P'_n P_k) \longrightarrow 0 \quad (x \rightarrow \pm 1),$$

because both polynomials and their derivatives stay finite. Therefore the weighted orthogonality theorem (4.2.20) applies directly:

$$\int_{-1}^1 P_n(x) P_k(x) dx = 0 \quad (n \neq k). \quad (4.2.23)$$

Thus the Legendre polynomials are orthogonal with weight $w(x) = 1$.

Example: Bessel equation and circular-drum modes

For a fixed nonnegative integer angular order m , the Bessel function $J_m(x)$ satisfies

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0. \quad (4.2.24)$$

For a radial mode on the unit disk, define

$$y(s) = J_m(\alpha s), \quad \alpha > 0, \quad 0 \leq s \leq 1.$$

The chain rule gives

$$\begin{aligned} y'(s) &= \alpha J_m'(\alpha s), \\ y''(s) &= \alpha^2 J_m''(\alpha s). \end{aligned}$$

Substituting these derivative factors into Bessel's equation at $x = \alpha s$ gives

$$s^2 y'' + sy' + (\alpha^2 s^2 - m^2) y = 0. \quad (4.2.25)$$

For $s > 0$, divide by s :

$$s y'' + y' + \left(\alpha^2 s - \frac{m^2}{s} \right) y = 0.$$

Example: Bessel equation and circular-drum modes (cont.)

Since

$$(sy')' = sy'' + y',$$

the equation becomes

$$(sy')' - \frac{m^2}{s}y + \alpha^2 sy = 0,$$

or

$$\frac{d}{ds} \left(s \frac{dy}{ds} \right) - \frac{m^2}{s}y = -\alpha^2 sy. \quad (4.2.26)$$

Thus

$$p(s) = s, \quad q(s) = -\frac{m^2}{s}, \quad w(s) = s, \quad \lambda = \alpha^2.$$

The endpoint $s = 0$ is singular, so the equation divided by s is interpreted on $(0, 1]$ together with regularity at the center.

For fixed m , let

$$y_i(s) = J_m(\alpha_i s), \quad y_j(s) = J_m(\alpha_j s).$$

Their eigenvalues are α_i^2 and α_j^2 . Applying (4.2.19) on $[\varepsilon, 1]$ with $p = w = s$ gives

$$[s(y_j y_i' - y_i y_j')]_{\varepsilon}^1 = -(\alpha_i^2 - \alpha_j^2) \int_{\varepsilon}^1 s y_i y_j ds. \quad (4.2.27)$$

Example: Bessel equation and circular-drum modes (cont.)

We must now check both endpoints; this is the part specific to the drum.

At $s = 1$, impose one common boundary condition on the whole family:

- Dirichlet: $y_i(1) = J_m(\alpha_i) = 0$ for every i ; or
- Neumann: $y_i'(1) = \alpha_i J_m'(\alpha_i) = 0$ for every i , where J_m' denotes differentiation with respect to the Bessel-function argument.

Either choice makes the boundary expression at $s = 1$ zero. For the Neumann condition with $m = 0$, $\alpha = 0$ also gives the constant mode $y(s) = 1$ with eigenvalue $\lambda = 0$; the positive α_i below label the nonconstant modes.

At $s = 0$, the regular Bessel solution has the expansions

$$J_m(z) = \frac{1}{m!} \left(\frac{z}{2}\right)^m + O(z^{m+2}) \quad (m \geq 1),$$

$$J_0(z) = 1 - \frac{z^2}{4} + O(z^4).$$

Hence, for $m \geq 1$,

$$y_i(s) = O(s^m), \quad y_i'(s) = O(s^{m-1}),$$

so

$$s(y_i y_i' - y_i' y_i) = O(s^{2m}) \longrightarrow 0.$$

Example: Bessel equation and circular-drum modes (cont.)

For $m = 0$, $y_i(s) = 1 + O(s^2)$ and $y_i'(s) = O(s)$, giving the same boundary limit, now of order $O(s^2)$. The singular Y_m solutions are excluded by regularity at the disk center.

Letting $\varepsilon \rightarrow 0^+$ in (4.2.27) and taking distinct positive α_i, α_j gives

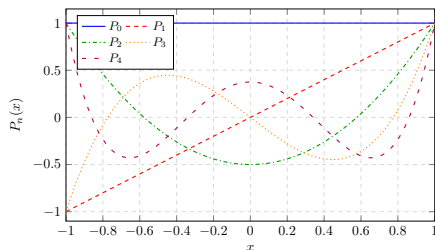
$$0 = -(\alpha_i^2 - \alpha_j^2) \int_0^1 s J_m(\alpha_i s) J_m(\alpha_j s) ds.$$

Since $\alpha_i^2 \neq \alpha_j^2$,

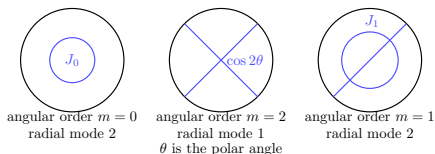
$$\int_0^1 J_m(\alpha_i s) J_m(\alpha_j s) s ds = 0 \quad (i \neq j). \quad (4.2.28)$$

The radial weight is therefore $w(s) = s$.

Legendre and Bessel modes



Legendre modes are orthogonal on $[-1, 1]$
with weight $w(x) = 1$.



Drum modes use the radial weight
 $w(s) = s$; a common boundary condition
at $s = 1$ selects the admissible α_i .

Putting a general equation into Sturm–Liouville form

Step 1: fully worked example—Hermite. Start from the concrete equation

$$y'' - 2xy' + \lambda y = 0. \quad (4.2.29)$$

Choose the integrating factor

$$Q(x) = \exp\left(\int_0^x -2t \, dt\right) = e^{-x^2}, \quad Q'(x) = -2xQ(x).$$

Multiplication of (4.2.29) by Q gives

$$Qy'' - 2xQy' + \lambda Qy = 0.$$

The first two terms combine by the product rule:

$$(Qy')' = Qy'' + Q'y' = Qy'' - 2xQy'.$$

Hence

$$(e^{-x^2} y')' = -\lambda e^{-x^2} y. \quad (4.2.30)$$

Reading off the data,

$$p(x) = e^{-x^2}, \quad q(x) = 0, \quad w(x) = e^{-x^2}.$$

Putting a general equation into Sturm–Liouville form (cont.)

The polynomial solutions occur at $\lambda_n = 2n$ and are the physicists' Hermite polynomials H_n , normalized to leading coefficient 2^n :

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2.$$

Chapter 6 derives these by series. For example, $H_2'' - 2xH_2' + 4H_2 = 8 - 16x^2 + 16x^2 - 8 = 0$. For nonnegative integers $n \neq m$, both H_n, H_m and their derivatives are polynomials, so

$$e^{-x^2} (H_n H_m' - H_n' H_m) \longrightarrow 0 \quad (x \rightarrow \pm\infty),$$

and $\lambda_n - \lambda_m = 2(n - m) \neq 0$. Weighted orthogonality therefore yields

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 0 \quad (n \neq m). \quad (4.2.31)$$

Step 2: general formula. Now abstract the calculation. Consider

$$p_0(x)u'' + p_1(x)u' + p_2(x)u = 0 \quad (4.2.32)$$

Putting a general equation into Sturm–Liouville form (cont.)

with real coefficients on an interval where $p_0(x) \neq 0$ and p_1/p_0 is continuous. Divide by p_0 :

$$u'' + \frac{p_1}{p_0} u' + \frac{p_2}{p_0} u = 0. \quad (4.2.33)$$

Define the integrating factor

$$Q(x) = \exp \left(\int_{x_0}^x \frac{p_1(t)}{p_0(t)} dt \right). \quad (4.2.34)$$

The chain rule gives $Q'/Q = p_1/p_0$. Multiplying (4.2.33) by Q ,

$$Qu'' + Q' u' + Q \frac{p_2}{p_0} u = 0.$$

The first two terms combine to $(Qu')'$, so

$$(Qu')' + Q \frac{p_2}{p_0} u = 0. \quad (4.2.35)$$

Putting a general equation into Sturm–Liouville form (cont.)

Thus the formally self-adjoint expression has

$$p = Q, \quad q = Q \frac{p_2}{p_0}.$$

Equivalently, without first dividing by p_0 , multiply the original equation (4.2.32) by Q/p_0 (not by Q alone).

If the spectral parameter occurs as

$$p_0 u'' + p_1 u' + (r_0(x) + \lambda r_1(x))u = 0, \quad (4.2.36)$$

the same steps give

$$(Qu')' + qu = -\lambda wu, \quad q = Q \frac{r_0}{p_0}, \quad w = Q \frac{r_1}{p_0}. \quad (4.2.37)$$

The positive-weight theory applies when r_1/p_0 has a fixed nonzero sign on the interval. If it is negative, replace λ by $-\lambda$ and w by $-w$. A sign-changing weight is not covered by that theory.

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- 1 Review of linear algebra
- 2 Sturm–Liouville operators
- 3 Green's function on an interval

A unit point source creates a jump

The machinery above solves the homogeneous problem $Lu = 0$. A *point source* turns it into a solution formula for $Lu = f$. On a regular finite interval $[a, b]$, let p be continuously differentiable with $p > 0$, let q be continuous, and take

$$L[u] = -(pu')' - qu,$$

the sign convention in which the eigenvalue problem reads $Lu = \lambda wu$, with separated self-adjoint boundary conditions on $[a, b]$.

Define the Dirac delta $\delta(x - x')$ operationally: it vanishes for $x \neq x'$ and, for continuous f ,

$$\int_a^b \delta(x - x') f(x) dx = f(x'), \quad a < x' < b.$$

It models a unit point source at x' , not an ordinary function.

The *Green function* $G(x, x')$ solves

$$L_x G = \delta(x - x')$$

under the problem's boundary conditions. Throughout this frame assume 0 is *not* an eigenvalue of L —equivalently, that $Lu = 0$ with these boundary conditions forces $u = 0$. This is what makes L invertible, so a unique Green function exists: it guarantees $W[y_L, y_R] \neq 0$ in the two-solution form below and $\lambda_n \neq 0$ in the modal series. If 0 is an

A unit point source creates a jump (cont.)

eigenvalue (as for the $m = 0$ Neumann drum of §4.2), the point-source problem is solvable only up to the null space and subject to a solvability condition, and both formulas must be modified. Away from x' the equation is homogeneous, so G is assembled from two solutions: y_L satisfying the left condition and y_R the right,

$$G(x, x') = \begin{cases} A y_L(x) y_R(x'), & x < x', \\ A y_L(x') y_R(x), & x > x', \end{cases}$$

with one shared amplitude A , because G must be continuous at x' : a jump in G would put a δ' into G'' , which the equation does not contain. Integrating $-(pG')' - qG = \delta$ over $[x' - \varepsilon, x' + \varepsilon]$ and letting $\varepsilon \rightarrow 0$ leaves the derivative jump

$$G'(x'^+, x') - G'(x'^-, x') = -\frac{1}{p(x')}. \quad (4.3.1)$$

Evaluating the jump from the piecewise form,

$$A[y_L(x')y_R'(x') - y_L'(x')y_R(x')] = A W[y_L, y_R](x') = -\frac{1}{p(x')},$$

A unit point source creates a jump (cont.)

so $A = -1/(p(x')W(x'))$. Abel's identity (Chapter 3) makes pW constant, so the value does not depend on where W is evaluated. The result is

$$G(x, x') = -\frac{y_L(x_{<})y_R(x_{>})}{p(x')W[y_L, y_R](x')}, \quad x_{<} = \min(x, x'), \quad x_{>} = \max(x, x'), \quad (4.3.2)$$

and $u(x) = \int_a^b G(x, x')f(x') dx'$ solves $Lu = f$ with the same boundary conditions. The jump condition (4.3.1) is local: it also applies when the interval is a circle (periodic angular coordinate), where the two solutions are matched periodically instead of by endpoint conditions. That jump is exactly what fixes modal coefficients when a field is expanded in eigenfunctions in a wave course.

The same function written in eigenfunctions

Alternatively expand G in the operator's own eigenfunctions. In addition to the invertibility assumption above, assume that these eigenfunctions form a complete basis in the weighted inner product. Let $L\phi_n = \lambda_n w \phi_n$ with real eigenfunctions and weighted norm $N_n = \int_a^b w \phi_n^2 dx$. Write $G(x, x') = \sum_n c_n \phi_n(x)$; then

$$\delta(x - x') = LG = \sum_n c_n \lambda_n w(x) \phi_n(x).$$

Multiplying by $\phi_m(x)$ and integrating, the weighted orthogonality gives $c_m \lambda_m N_m = \phi_m(x')$, hence

$$G(x, x') = \sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(x')}{\lambda_n N_n}. \quad (4.3.3)$$

(For complex modes the conjugated orthogonality replaces $\phi_n(x')$ by $\overline{\phi_n(x')}$.) Equating (4.3.3) with (4.3.2) is the link between a *modal series* and the *two-solution* form of the same Green function.

The same function written in eigenfunctions (cont.)

Conversely, the eigenfunction expansion $f(x) = \sum_n c_n \phi_n(x)$ with $c_n N_n = \int_a^b w(x') \phi_n(x') f(x') dx'$ can be rewritten with the bracket $\sum_n \phi_n(x) \phi_n(x') / N_n$; that bracket reproduces the delta source in the weighted inner product:

$$\sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(x')}{N_n} = \frac{\delta(x - x')}{w(x)}. \quad (4.3.4)$$

This *completeness relation* is the precise sense in which the eigenfunctions span the space. A modal field expansion in a later wave course is exactly this statement with the weight and basis chosen by the geometry. The equality is understood through integration against an admissible function; it does not claim that the delta distribution is an ordinary pointwise function.

Chapter summary

- Functions behave as vectors under $(u, v) = \int_a^b u \bar{v} dx$, or its weighted version. Our convention is linear in the first slot: $(cu, v) = c(u, v)$, $(u, cv) = \bar{c}(u, v)$.
- Orthogonality gives coefficients by projection: $a_n = (f, u_n)/\|u_n\|^2$. For $u_n = \sin(n\pi x)$ on $(0, 1)$, $\|u_n\|^2 = 1/2$.
- Integration by parts identifies both the formal adjoint and the boundary term. The expression $p_0 u'' + p_1 u' + p_2 u$ with real coefficients is formally self-adjoint when $p_1 = p_0'$, giving $(pu')' + qu$.
- Boundary conditions must make $[p(u\bar{v}' - u'\bar{v})]_a^b$ vanish. Under these conditions and a positive weight, $(pu')' + qu = -\lambda wu$ has real eigenvalues and weighted-orthogonal eigenfunctions for distinct eigenvalues.
- Multiplying a general second-order equation by an integrating factor puts its derivative terms into $(Qu')'$, with $Q = \exp \int (p_1/p_0) dx$. The Legendre, Bessel, and Hermite weights are 1, s , and e^{-x^2} .
- For $L = -(pu')' - qu$, the Green function solves $LG = \delta(x - x')$: it is continuous at x' , its derivative jumps by $-1/p(x')$, and it can be written either as $-y_L(x_<)y_R(x_>)/(pW)$ (4.3.2) or as the modal series (4.3.3); the basis also satisfies the completeness relation (4.3.4).